

This document is exploratory and is not yet linked from the main project index (README.md or GUIDE.md). It covers the ZP-J valuation bridge work (ZPJ_Scale, ZPJ_SelfApp, ZPJ_AczelConn, ZPJ_Model) not covered in the main ZP-J companion. For the main project, see README.md.

ZP-J Valuation Bridge — Exploratory Companion

Why $\perp$ Is the Only Fixed Point

Deriving Self-Reference from Arithmetic

ZP Companion | Version 1.1 | May 2026

This companion explains the ideas in plain language. It covers the ZP-J valuation bridge — four Lean sub-files that derive AFA self-containment from valuation arithmetic. It is exploratory and not a formal document.

What This Document Does

ZP-J proved T-EXEC: the bottom element $\perp$ of any ZP-A semilattice with AFA grounding is the unique Quine atom ($\perp = \{\perp\}$). But T-EXEC imported three facts from AFA as class fields — axioms in the Lean encoding: selfMem (self-membership as a proposition), bot_self_mem ($\perp$ is self-containing), and quine_unique (any two self-containing elements are equal). These appear as axioms because the abstract lattice lacks the machinery to derive them from anything more primitive.

The four ZP-J sub-files ask the next question: can those AFA class fields be derived from something more basic? The answer is yes — from a valuation structure. If a scale operation strictly increases a measure, then $\perp$ (the element of infinite measure) is forced to be the unique fixed point of scale. The specific AFA content needed here — self-membership, $\perp$ as the unique self-containing element, and uniqueness — follows as a theorem, not an assumption.

The Valuation Argument

A valuation assigns a numeric measure to each element of the lattice. The bottom element $\perp$ has measure ∞ — represented as $\top$ (the top element of $\mathbb{N}\infty$, the extended naturals). Every other element has a finite measure. Elements with higher measure are structurally closer to $\perp$ in the valuation; $\perp$ itself sits at measure ∞ . Formally: $\text{val}(\perp) = \top$, and $\text{val}(x) \neq \top$ for $x \neq \perp$.

The scale operation moves each element one step further in the valuation — strictly increasing the measure. Formally: $\text{val}(\text{scale } x) = \text{val } x + 1$ for every $x \neq \perp$. This single axiom has one powerful consequence: scale cannot have a fixed point anywhere other than $\perp$.

The proof is a one-line contradiction. Suppose $\text{scale } x = x$ and $x \neq \perp$. Then $\text{val}(x) = \text{val}(\text{scale } x) = \text{val}(x) + 1$. But $\text{val}(x)$ is finite (since $x \neq \perp$), and no finite number equals itself plus one. Contradiction. So the only fixed point of scale is $\perp$. (The exception " $x \neq \perp$ " is intentional: at $\perp$ itself, scale fixes $\perp$ by axiom `scale_bot`, and $\infty + 1 = \infty$ in $\mathbb{N}\infty$ is consistent — the $+1$ rule characterises non-bot elements, not $\perp$ itself.)

The impossible equation

In the natural numbers $\mathbb{N}$: if $n = n + 1$, what is n ? There is no solution. Adding 1 always moves you forward — you can never stay in place. Only at the extended point $\top$ in $\mathbb{N}_\infty$ does this equation have a solution: $\top + 1 = \top$ in $\mathbb{N}_\infty$ (successor saturates at $\top$ — WithTop arithmetic by convention). The valuation argument is the same: $\text{val}(\text{scale } x) = \text{val } x + 1$ means scale always moves forward in the measure. The only escape is already being at measure ∞ — which means $x = \perp$.

The Derivation Chain

The valuation argument runs through three layers. Each step takes the output of the previous one and derives something stronger:



Sorry-free derivation chain. Each arrow replaces an AFA axiom with a theorem proved from valuation arithmetic alone.

Three-step sorry-free derivation. `ValuationStructure` provides the arithmetic input. `AbstractSelfApp` extracts the minimal fixed-point structure. `AFA Content` is the output: self-membership, $\perp$ as the unique self-containing element.

Step 1 — `ValuationStructure` → `AbstractSelfApp`. `ZPJ_Scale.lean` builds the instance to `AbstractSelfApp`: set `selfApp = scale`, then derive `fixed_bot` from `scale_bot` and `unique_fp` from the one-line valuation contradiction above. `AbstractSelfApp` has just two fields: a self-application operation and the claim that $\perp$ is its unique fixed point.

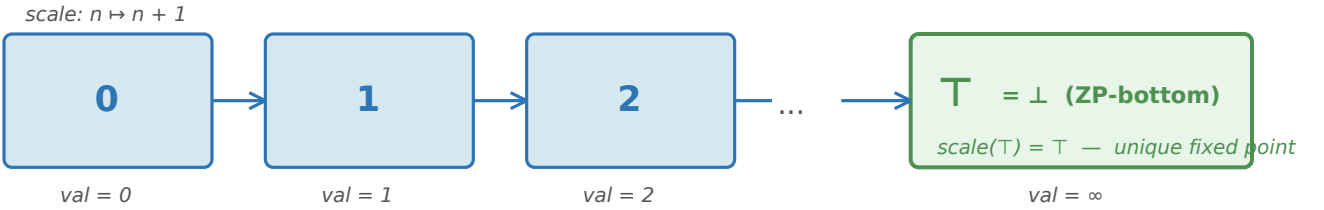
Step 2 — `AbstractSelfApp` → `AFA Content`. `ZPJ_SelfApp.lean` derives from `fixed_bot` and `unique_fp` alone: `selfMem x := (selfApp x = x)`; `bot_self_mem` ($\perp$ is self-containing, from `fixed_bot`); `quine_unique` (any two self-containing elements are equal, from `unique_fp`); and `{x | selfMem x} = {⊥}` — the self-containing set has exactly one element.

All steps are sorry-free. Every result depends only on `[propext, Classical.choice, Quot.sound]` — no AFA axioms are imported. The specific ZP formal predicates that model AFA self-membership — `selfMem`, `bot_self_mem`, `quine_unique`, and the singleton theorem — are derived from the arithmetic of `val` and `scale`. This is not a derivation of AFA's Anti-Foundation Axiom system, which is a richer structure; it is a derivation of the specific fixed-point content ZP uses to model self-containment.

The Concrete Model: $\mathbb{N}_\infty$

The derivation chain works for any type carrying a `ValuationStructure`. `ZPJ_Model.lean` provides the concrete witness: $\mathbb{N}_\infty$ (the extended natural numbers, `WithTop ℕ` in Lean 4) with a specific choice of join, bottom, scale, and valuation.

`ZPSemilattice` uses a reversed order on $\mathbb{N}_\infty$. In the ZP partial order, $x \leq_{\text{ZP}} y$ iff $\min x y = y$ iff $x \geq y$ in $\mathbb{N}_\infty$'s natural order. Under this reversed order, min becomes the join (least upper bound in the ZP lattice), and $\top$ — the natural maximum of $\mathbb{N}_\infty$ — becomes the ZP-bottom. So: join = min, bot = $\top$, val = identity function (val $n = n$, val $\top = \top$), scale = successor ($n \rightarrow n + 1$).



$\mathbb{N}_\infty$ with join = min and bot = $\top$: the ZP order reverses $\mathbb{N}$ natural order. $\top$ is the unique fixed point of scale — ZP-bottom ($\perp$).

$\mathbb{N}_\infty$ with scale = successor. Each $n \in \mathbb{N}$ has val $n = n$; scale moves n to $n+1$. $\top$ is the unique fixed point: $\top + 1 = \top$ in $\mathbb{N}_\infty$ (WithTop arithmetic). $\top$ is ZP-bottom ($\perp$).

The theorem `natInf_selfMem_singleton` states: $\{x : \mathbb{N}_\infty \mid x + 1 = x\} = \{\top\}$. This is AFA content derived entirely from arithmetic: the "self-containing set" in this model is exactly $\{\top\} = \{\perp\}$. The proof delegates to `val_selfMem_singleton` from `ZPJ_Scale.lean` — the same abstract result that applies to any `ValuationStructure`.

The ZP order on $\mathbb{N}_\infty$ is reversed: $\top$ is the bottom (most open, val = ∞), and 0 is the maximum (fully constrained, val = 0). Scale moves elements toward $\top$ — toward less constraint, toward the bottom. This is the "indexed family" picture: each level n is a state of constraint, and $\top$ is the unreachable limit where all valuation chains converge.

The 2-Adic Parallel

The valuation structure was motivated by the 2-adic integers $\mathbb{Z}_2$. In $\mathbb{Z}_2$: scale $x = 2 \times x$, and val $x = v_2(x)$ — the 2-adic valuation, the exponent of the highest power of 2 dividing x . The 2-adic valuation assigns infinite measure to 0 by convention: $v_2(0) = \infty$. This is consistent with the fact that 2^n divides 0 for every n , but the infinitary value is part of the definition of the extended 2-adic valuation, not derived from divisibility alone.

The key identity: $v_2(2x) = v_2(x) + 1$ for $x \neq 0$. Scale ($\times 2$) strictly increases the 2-adic valuation by exactly 1 — the same axiom as `val_scale` in `ValuationStructure`, now in $\mathbb{Z}_2$. The unique fixed point of $\times 2$ is 0, proved by ring arithmetic: $2x = x$ gives $x = 2x - x = 0$ in $\mathbb{Z}_2$.

$\mathbb{Z}_2$ is not a `ZPSemilattice` (it is a ring, not a join-semilattice), so it cannot be a formal `ValuationStructure` instance. `ZPJ_Scale.lean` §V proves all four `ValuationStructure` axioms for $\mathbb{Z}_2$ as standalone theorems — sorry-free, using Mathlib's `PadicInt` library. The $\mathbb{N}_\infty$ model provides the formal instance; $\mathbb{Z}_2$ shows the same abstract structure in a mathematically rich concrete object.

Why $\mathbb{Z}_2$ and not $\mathbb{R}$?

In $\mathbb{R}$, the only compatible valuation is the trivial one ($v(x) = 0$ for all $x \neq 0$, $v(0) = \infty$), which gives no useful scale operation: $v(2x) = v(x)$ for all $x \neq 0$. There is no canonical non-Archimedean valuation on $\mathbb{R}$ satisfying $v(2x) = v(x) + 1$. $\mathbb{Z}_2$ carries exactly such a valuation — one where 0 has infinite measure and multiplication by 2 shifts every other element one step up. This is why the fixed-point argument works in $\mathbb{Z}_2$ and not in $\mathbb{R}$.

DC-Free: Identification Replaces Construction

Aczel's original proof that the largest fixed point of a set-continuous operator Φ is $J_\Phi = \bigcup \{x \mid x \subseteq \Phi(x)\}$ used the Axiom of Dependent Choice (DC) to construct an ω -chain (Non-Well-Founded Sets, 1988, ch. 6, p. 77). Aczel noted explicitly: "I do not know if this use of the axiom of dependent choices was essential."

ZPJ_AczelConn.lean answers this for the self-membership case: DC is not needed. quine_unique — AFA's uniqueness clause — directly forces any self-containing element to equal $\perp$. If x is self-containing and $\perp$ is self-containing, then $x = \perp$ in one step. No chain needed. Identification replaces construction.

The valuation bridge has the same structure. unique_fp says: if scale $x = x$ then $x = \perp$. The fixed point is identified by the one-line valuation contradiction — not by constructing a sequence converging to it. DC-free. This matters because DC is a non-trivial existence principle. DC-free proofs work in settings where choice is restricted, and formally they carry a lighter axiom footprint.

Scope: ZPJ_AczelConn.lean proves DC is unnecessary for the self-membership case specifically. Whether it can be eliminated for all set-continuous operators Φ remains open. One natural conjecture — not established in the literature — is that DC-eliminability depends on whether uniqueness holds for the specific Φ . ZP provides a positive answer for the case ZP cares about.

Key Results — ZP-J Valuation Bridge

ValuationStructure (ZPJ_Scale.lean): $\text{val}(\text{scale } x) = \text{val } x + 1$ forces $\perp$ to be the unique fixed point of scale — AFA content derived, no AFA axioms imported. ✓ AbstractSelfApp (ZPJ_SelfApp.lean): minimal fixed-point typeclass from which selfMem, bot_self_mem, quine_unique, $\{x \mid \text{selfMem } x\} = \{\perp\}$ are theorems. ✓ $\mathbb{N}_\infty$ model (ZPJ_Model.lean): concrete ValuationStructure — natInf_selfMem_singleton : $\{x : \mathbb{N}_\infty \mid x+1=x\} = \{\top\}$. ✓ 2-adic parallel (ZPJ_Scale.lean §V): q2Val_scale proved sorry-free from Mathlib PadicInt. ✓ DC-free (ZPJ_AczelConn.lean): identification replaces construction for self-membership. ✓ All results: [propext, Classical.choice, Quot.sound] — no sorryAx anywhere.